

Question_3:

Code:

```
1  /*
2  QUESTION_3:
3  Insert the given elements into a Circular Linked List in sorted order.
4  The final Circular Linked List must contain the elements
5  in ascending order.
6  */
7  #include <iostream>
8  using namespace std;
9
10 struct Node {
11     int data;
12     Node *next;
13 };
14 void sortedInsert(Node *&head, int x) {
15     Node *p = new Node{x, NULL};
16
17     if (!head) {
18         head = p;
19         p->next = head;
20         return;
21     }
22     if (x <= head->data) {
23         Node *t = head;
24
25         while (t->next != head)
26             t = t->next;
27         p->next = head;
28         t->next = p;
29         head = p;
30         return;
31     }
32     Node *t = head;
33
34     while (t->next != head && t->next->data < x)
35         t = t->next;
36     p->next = t->next;
37     t->next = p;
38 }
```

```
39  int main() {
40      int n, x;
41      cin >> n;
42
43      Node *head = NULL;
44      for (int i = 0; i < n; i++) {
45          cin >> x;
46          sortedInsert(head, x);
47      }
48      Node *p = head;
49
50      for (int i = 0; i < n; i++) {
51          cout << p->data << " ";
52          p = p->next;
53      }
54      return 0;
55  }
```

Output:

Test Case_1:

```
6
10 20 30 45 34 56
10 20 30 34 45 56
PS C:\Users\Prabin Yadav\
```

Test Case_2:

```
6
1 2 11 12 56 90
1 2 11 12 56 90
PS C:\Users\Prabin Yadav\
```